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Secretory leukocyte protease inhibitor mediates non-redundant functions necessary for normal wound healing

GILLIAN S. ASHCROFT[1], KEJIAN LEI[1], WENWEN JIN[1], GLENN LONGENECKER[2], ASHOK B. KULKARNI[2], TERESA GREENWELL-WILD[1], HOLLIE HALE-DONZE[1], GEORGE MCGRADY[1], XIAO-YU SONG[1] & SHARON M. WAHL[1]

[1]*Oral Infection and Immunity Branch,* [2]*Gene Targeting Facility, National Institute of Dental & Craniofacial Research, Building 30, 30 Convent Drive, MSC 4352, National Institutes of Health, Bethesda, Maryland 20892, USA*

G.A. & K.L. contributed equally to this study.

Correspondence should be addressed to S.W.; email: smwahl@dir.nidcr.nih.gov

Secretory leukocyte protease inhibitor (SLPI) is a serine protease inhibitor with anti-microbial properties found in mucosal fluids. It is expressed during cutaneous wound healing. Impaired healing states are characterized by excessive proteolysis and often bacterial infection, leading to the hypothesis that SLPI may have a role in this process. We have generated mice null for the gene encoding SLPI (*Slpi*), which show impaired cutaneous wound healing with increased inflammation and elastase activity. The altered inflammatory profile involves enhanced activation of local TGF-β in *Slpi*-null mice. We propose that SLPI is a pivotal endogenous factor necessary for optimal wound healing.

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Secretory leukocyte protease inhibitor (SLPI) is a protein of 12 kDa composed of 2 cysteine-rich domains with a protease inhibitory site situated at leucine 72 in the carboxy-terminal domain[1,2]. SLPI is a potent inhibitor of serine proteases, such as neutrophil elastase, cathepsin G, mast cell chymase and a chymotrypsin-like enzyme found in the stratum corneum[3–6]. In addition to its primary function as an anti-protease, SLPI has also been shown to have multiple functions relevant to innate host defense, including anti-microbial and anti-inflammatory activity, the control of intracellular enzyme synthesis, and the suppression of monocyte matrix metalloproteinase production and activity[7–11]. SLPI was first described in seminal fluid and is predominantly found bathing mucosal surfaces in bronchial fluids, cervical fluids and saliva, but can also be detected in rodent inflammatory cells and in human and rodent dermal keratinocytes[11–16]. Because its systemic levels are low, SLPI is thought to act at a tissue level, unlike circulating serine protease inhibitors such as α1-antitrypsin and α2-macroglobulin. Moreover, the recent discovery of epithelial SLPI expression during cutaneous wound repair has generated interest in the role of this molecule in the normal wound-healing response[14].

A number of documented functions of SLPI *in vitro* suggest a potential role during wound healing. Beyond its anti-inflammatory, anti-viral, anti-fungal and anti-bacterial properties, SLPI antagonizes lipopolysaccharide (LPS) -induced pro-inflammatory mediator synthesis by monocytes and macrophages[7,8,10,11,13,16]. When viewed in the context of a chronic non-healing wound, in which inflammation and bacterial infection contribute to delayed healing, SLPI is a potential candidate for enhancing the healing response; however, the major function of SLPI as a local anti-elastase agent is the most intriguing. Proteinases derived from polymorphonuclear leukocytes (PMNs) such as elastase and cathepsin G are involved in extensive pathological tissue destruction in a number of disorders, including delayed and chronic wound healing[17,18]. An abnormal balance of serine protease activity versus endogenous inhibitor has been shown to be of potential importance not only in human chronic non-healing wounds, but also in age-related impaired acute wound healing[18–20]. One explanation for this is the ability of elastase to degrade a variety of structural and functional proteins deposited in wounds (such as collagen and fibronectin), as well as key growth factors such as TNF-α (refs. 21–23). Moreover, elastase indirectly enhances matrix degradation through its ability to activate matrix metalloproteinases[11]. Despite the strong association between abnormal elastase activity and impaired wound healing, a causal effect has never been proven, and no useful animal model of a chronic wound currently exists. To this end, we generated animals deficient for the elastase inhibitor SLPI, which revealed a complex interaction of proteases and inflammatory mediators involved in aberrant wound healing. Thus, we not only document an essential role for SLPI in multiple regulatory pathways during wound healing, but also provide a new animal model for impaired wound healing.

Results

Generation of *Slpi*-mutant animals

To directly assess the role of SLPI in the wound healing process *in vivo*, we generated a mouse null for *Slpi* by targeted recombination (K.L. *et al.*, manuscript in preparation). A targeting vector was constructed in which the 849-bp promoter region and the 46-bp exon I of *Slpi* were replaced with a neomycin-resistance gene and introduced into R1 embryonic stem (ES) cells. We injected targeted ES clones into blastocysts from C57BL/6 mice and transferred them to B6D2F1 foster mothers. We identified homozygous mutant progeny by PCR and Southern-blot analysis of tail genomic DNA, and northern-blot analysis confirmed the absence of SLPI expression from *Slpi*-null mice tissues (data not shown). SLPI, localizing to mucosal epithelial cells in human tissues, was similarly detected at the RNA level in mouse mucosae. SLPI was also identified at low levels in skin and lymphoid or-

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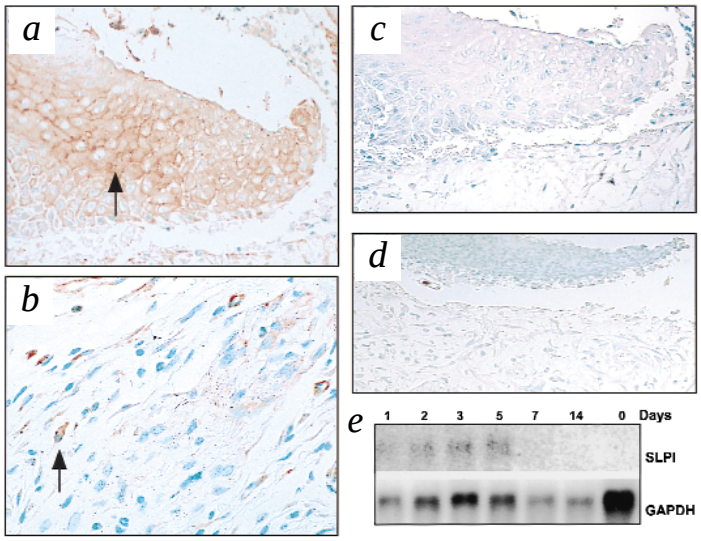


Fig. 1 SLPI is expressed during normal wound repair. SLPI protein was observed during wound healing in the suprabasal epidermal cells at the migrating tip (***a***, arrow pointing to suprabasal cells; , day 3 wound) and in wild-type inflammatory cells (***b***, arrow; , day 3 wound) in granulation tissue. Magnification, ×20. ***c***, Absence of staining on parallel sections using the isotype control antibody (magnification, ×10). ***d***, Day 3 cutaneous wound from the *Slpi*-null mouse does not stain with anti-SLPI antibody (magnification, ×10). ***e***, Increased *Slpi* mRNA expression was seen from day 1, peaking at day 5 post-wounding in wild-type mice. By day 7, SLPI expression decreased markedly.

gans in wild-type mice (data not shown), consistent with its identification in rodent macrophages[11,16]. *Slpi*-null mice had normal fetal development, survived into adulthood and were fertile, and showed no obvious pathological phenotype.

SLPI expression during normal wound healing

We assessed the temporal and cellular source of SLPI expression during normal wound healing in wild-type C57BL/6 mice following full-thickness incisional wounds excised at days 1 to 14 post-wounding. Although not apparent in normal epidermis, we detected SLPI staining in the suprabasal layers in the migrating epidermis (Fig. 1). Inflammatory cell staining, in contrast to human skin injury[14], was also evident from day 1 post-wounding, persisting until day 7 post-wounding (Fig. 1*b*). No staining for SLPI was evident in *Slpi*-null mouse skin before or after wounding (Fig. 1*d*). Moreover, by northern-blot analysis of wound tissue, *Slpi* RNA levels were apparent within the first day post-wounding, increasing through day 5 and then decreasing to baseline by day 7 (Fig. 1*e*), consistent with the immunohistochemical pattern. The temporal increase in SLPI expression during early wound healing implicated a role for this peptide in inflammation and/or repair.

Effects of *Slpi* deletion on wound healing

Following full-thickness dorsal incisional wounding of mice deficient in SLPI, an impaired rate of wound healing was apparent (Fig. 2*a*). Within the first day, but most notable on the third day after wounding (when closure was apparent in wild type), *Slpi*-null animals had enlarged wounds, the margins of which failed to close. Wound areas were increased in the *Slpi*-null mice macroscopically and microscopically (Fig. 2), and quantification of the cross-sectional wound area from day 3 post-wounding revealed up to 900% increase up to day 14 post-wounding (Fig. 2*b*; $P < 0.05$ on days 3, 5 and 14). In addition, there was a substantial increase in cells per unit area reflecting a massive local accumulation of leukocytes (Fig. 3*a–e*; $P < 0.05$ on days 1, 3 and 5). Giemsa staining in conjunction with immunostaining using a macrophage-specific antibody indicated that neutrophils and monocytes were increased in the null wounds compared with wild-type within the first 24 hours (Fig. 3*a–e*). We saw no differences in circulating peripheral leukocyte counts in wild-type and null mice (data not shown), indicating that the massive accumulation of leukocytes in the wounds of the *Slpi*-null mice was not due to systemic differences in hematopoiesis. Additionally, leukocytes derived from *Slpi*-null bone marrow appeared functionally competent as assessed by phagocytosis and oxidative burst activity, required for defense against microbial colonization (data not shown).

Elastase activity is increased in *Slpi* wounds mutants

The expansive wounds in *Slpi*-deficient mice suggested unresolved inflammation, a failure of re-epithelialization, an inability to generate matrix and/or an inability to counter local proteolysis by serine proteases. Because neutrophil elastase has been implicated in matrix degra-

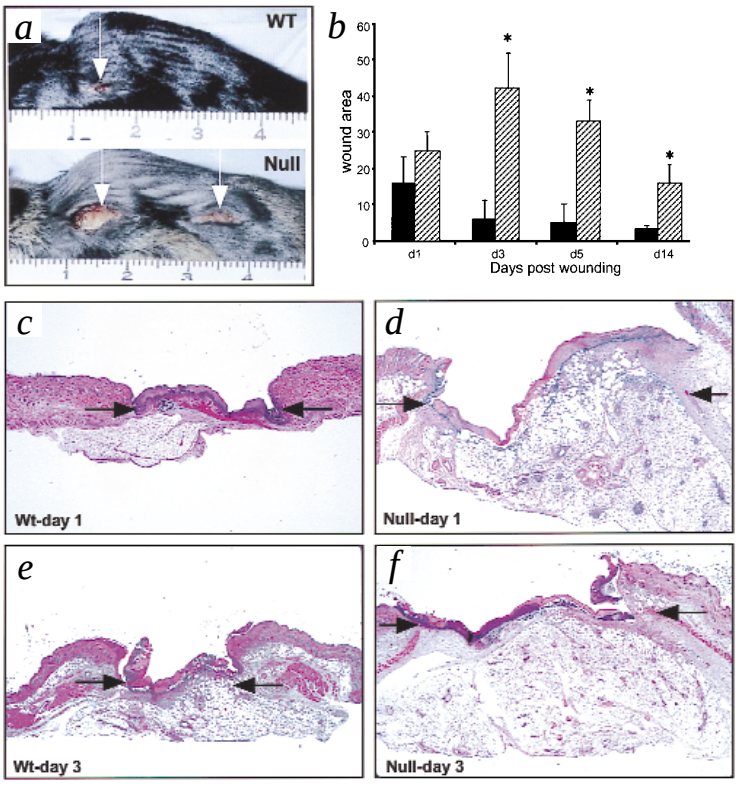


Fig. 2 Delayed wound healing in *Slpi*-null mice. ***a***, Macroscopically, the wounds in null mice (bottom) were larger than those of wild-type(WT). Day 3 wounds are shown. ***b***, Wound areas in wild-type(■) and null(▨) mice(were determined (measured below the clot and above the panniculus muscle) using an Optimas program [38]. Results are mean ± s.e.m., $n = 10$, for each time-point and group. *$P < 0.05$ compared with wild type. ***c–f***, Wounds were wider in null mice, with an increased inflammatory response compared with that of wild-type. Arrows demarcate wound margins. Magnification, ×2.5.

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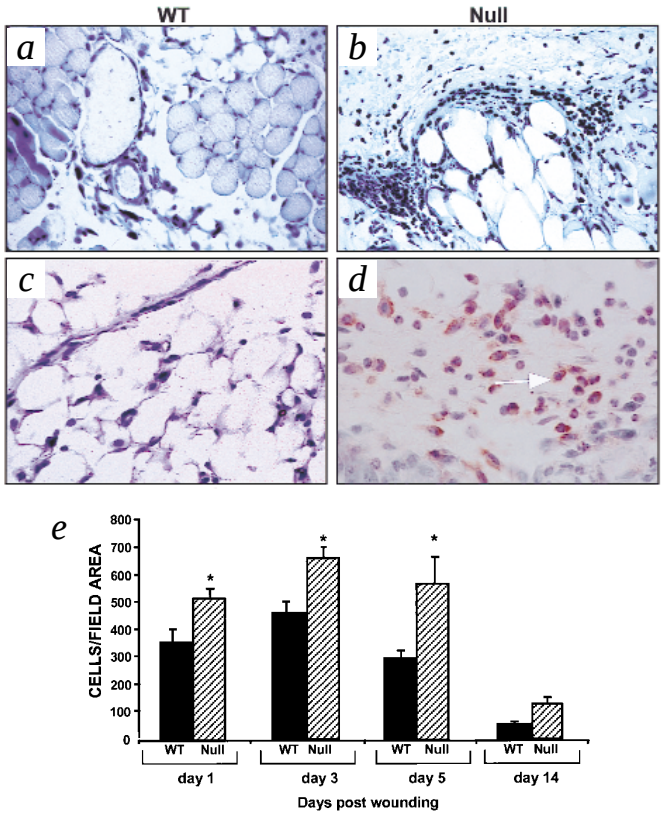


Fig. 3 Increased inflammation in the *Slpi*-null wounds. ***a***,***b***, Giemsa staining illustrates increased inflammatory cells in null compared with wild-type day 3 wounds. ***c***,***d***, Mac-2 staining identifies increased number of macrophages(arrow) in null compared with wounds. Magnification, ×40. ***e***, Cell numbers per unit area were quantified at days 1–14 post-wounding. Results are means ± s.e.m., $n = 10$, for each time point and group. *$P < 0.05$ compared with wild-type.

dation associated with impaired wound healing[18,20], we initially focused on the effects of *SLPI* deletion on local elastase activity. No elastase activity was observed in protein extracts from normal skin from either phenotype. Wound elastase activity was significantly increased from day 3 to day 14 post-wounding in the wounds of null mice compared with wild type (Fig. 4*a*; $P < 0.05$ on days 3, 5 and 14 post-wounding), reflecting increased cell numbers and/or a failure of elastase inhibition. One potential consequence of this raised proteolytic activity was a decrease in collagen matrix deposition in the wounds of the null mice as determined by both immunohistochemical and histological techniques (Fig. 4*b*–*e*). Moreover, the observations that elastase activates matrix metalloproteinases[24] (MMPs), coupled with the SLPI-mediated suppression of monocyte MMP synthesis and MMP activity[10,11], suggest that the absence of SLPI and the increased local levels of elastase maximize collagen degradation and lead to impaired healing.

To test the premise that impaired wound healing was a direct result of the absence of SLPI, we applied exogenous SLPI to cutaneous wounds to determine whether the pathology was reversible. Application of SLPI over the course of two days resulted in a reversal of the wound-healing phenotype, with narrower wounds and reduced inflammation (Fig. 5; wound areas in *Slpi*-null mice, 28 ± 1 versus 4 ± 2 after SLPI treatment; $P = 0.007$). Enhanced healing occurred in response to SLPI treatment in wild type, but it was not significant. Associated with the accelerated healing response in the null mice was a reduction in proteolytic activity. Elastase was reduced compared with untreated wounds from day 3 post-wounding (wild type, 10 ± 5 ng/ml elastase activity; null, 42 ± 4; wild type treated with SLPI, 9 ± 4; null treated with SLPI, 25 ± 10; $P = 0.05$).

Effects of SLPI deletion on the expression of cytokines

As the absence of SLPI was associated with an increased inflammatory cell influx that was dampened by exogenous SLPI, we investigated the mechanisms underlying the exuberant inflammatory response. If circulating inflammatory cells are to infiltrate the site of injury, they must first respond to a chemoattractant signal, and adhere to and traverse the endothelial barrier[25]. Inflammatory cells are attracted to the wound site by multiple stimuli including TGF-β (which has been shown to be expressed during the wound-healing process in a distinct temporal pattern[25,26]), chemokines and other cytokines.

Direct evidence supports the role of SLPI in modulating cytokine and chemokine production through its ability to suppress NF-κB activation *in vitro*[16] and *in vivo*[11,27]. Analysis of mRNA encoding TGF-β1, a powerful chemoattractant, revealed

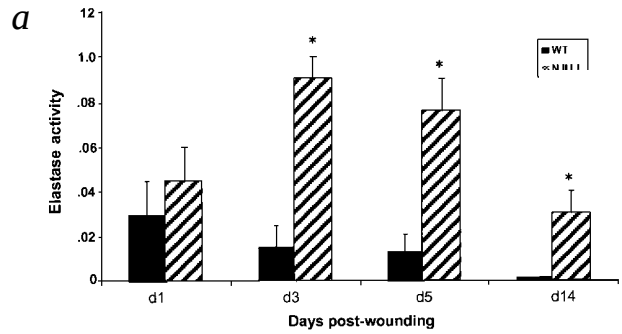
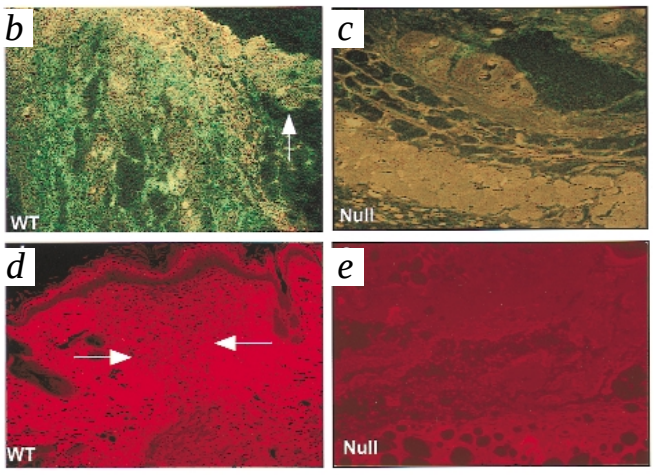


Fig. 4 The absence of SLPI leads to increased wound elastase activity and reduced matrix. ***a***, Elastase activity is increased from day 3 to 14 in null(▨) compared with wild-type(■)wound tissue. Results are expressed as μg/ml elastase activity/20 μg wet weight, and are means ± s.e.m., $n = 5$, for each group and time point. *$P < 0.05$ compared with wild-type. ***b***,***c,*** Collagen immunostaining (green) at day 3 was reduced in the null wounds. Arrow represents epidermis. Magnification, ×20. ***d,e***, Picrosirius red staining confirms the reduction in wound collagen in the *Slpi*-null day 7 wounds. Arrows represent wound margins. Magnification, ×20.



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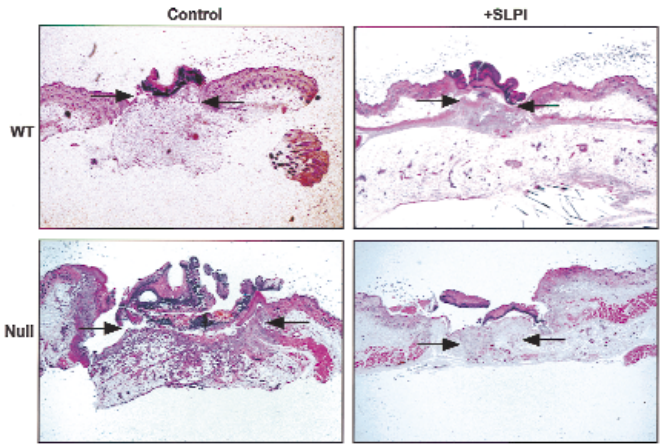


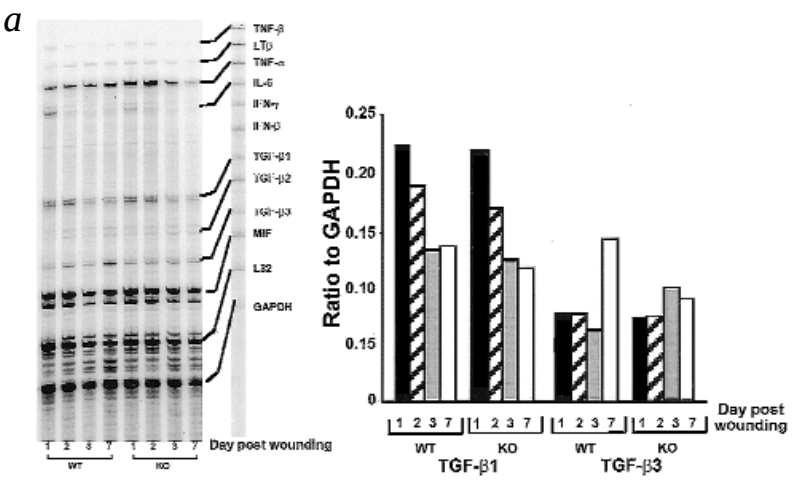
Fig. 5 Exogenous SLPI reverses the impaired wound healing phenotype in *Slpi*-null mice. At day 5 (control, no treatment), wounds in the wild-type (WT) mice were narrower (Arrows indicate wound margins.) with increased matrix compared with null mice. When SLPI was injected before wounding and at day 1 (+SLPI), more rapid healing was observed with reduced inflammation(Arrows indicate wound margins.). Magnification, ×2.5.

highest expression on day 1, but this was not significantly different between wild-type or null wounds (Fig. 6*a*). Increased levels of TGF-β3 RNA in wild-type day 7 wounds, concurring with previous studies[26] were not observed in the null wounds. This is consistent with evidence that upregulation of this isoform occurs after completion of the proliferative phase of wound healing[28], which is a delayed event in the null wounds.

Constitutive activation of TGF-β in the absence of SLPI

Although we detected no differences in TGF-β1 mRNA or total TGF-β protein (Fig. 6) in control and null wounds, we found an increase in constitutively active TGF-β in the wounds of *Slpi*-null mice (Fig. 6*b*). Both clot and inflammatory cells stained positively for active TGF-β (Fig. 6*b*,*c*). Moreover, following treatment of the wounds with exogenous SLPI, a decrease in active TGF-β staining was apparent (Fig. 6*c*). These findings provide an underlying mechanism by which excessive leukocyte recruitment and inflammation may occur in the absence of SLPI. Numerous studies have emphasized that the regulation of TGF-β and its chemotactic activity is multi-factorial and is dependent on local factors, including protease-dependent release from the matrix pool and cleavage to the active form, and a protease-independent requirement for cellular activation[29–32]. As a serine protease inhibitor, SLPI might be involved in blocking plasmin-dependent activation of latent TGF-β. SLPI, however, did not block plasmin activation of TGF-β or urokinase plasminogen activator (uPA) cleavage of plasminogen in *in vitro* assays (data not shown). Because latent TGF-β binding protein (LTBP) binds to dermal elastin fibers and is released by elastase, it is likely that the unchecked elastase activity in *Slpi*-null animals indirectly facilitates the activation of TGF-β in a protease-dependent manner[33–36].

Another potential mechanism for the action of SLPI might involve suppression of macrophage activation[11], leading to impaired downstream cellular activation of TGF-β. SLPI is an endogenous inhibitor of NFκB; thus it is conceivable that the lack of this molecular brake may result in enhanced macrophage activation[11,27,37]. Consistent with this possibility, a sustained increase in NFκB activation was observed in LPS-activated *Slpi*-null macrophages, paralleled by a reduction in IκB protein (data not shown). These data indicate that SLPI mediates its effects by both intracellular and extracellular pathways, consistent with the pronounced inflammation and impaired healing that occurs in its absence. To document a role for SLPI in the macrophage activation process leading to activation of latent TGF-β, we added SLPI to LPS-stimulated macrophages in a defined kinetic pattern and monitored the release of biologically active TGF-β (ref. 32). Whereas SLPI specifically inhibited TGF-β activation when added to the cells before, or



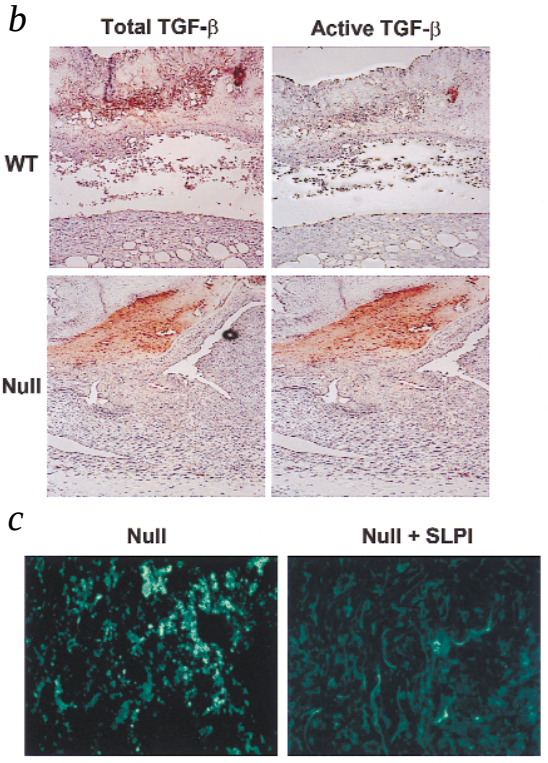


Fig. 6 TGF-β activation, but not RNA, is increased in the absence of SLPI. ***a***, Total RNA was pooled (n = 10) for the RNase protection assay. For quantification, band densities were normalized to GAPDH using an image analysis program (ImageQuant, Molecular Dynamics). An increase in TGF-β1 mRNA expression was observed at days 1 and 2 in both phenotypes. Data are representative of two similar RPAs. ***b***, Day 2 wounds stained for total or active TGF-β show an increase in active TGF-β in *Slpi*-null mice, mainly staining the clot, but also sub-epidermally. Secondary antibody controls were negative. ***c***, At day 3 post-wounding, active TGF-β was localized to wound inflammatory cells, with increased numbers in the null wounds. Following exogenous SLPI treatment, a decrease in cellular staining for active TGF-β was observed in the null wounds. Magnification, ×20.

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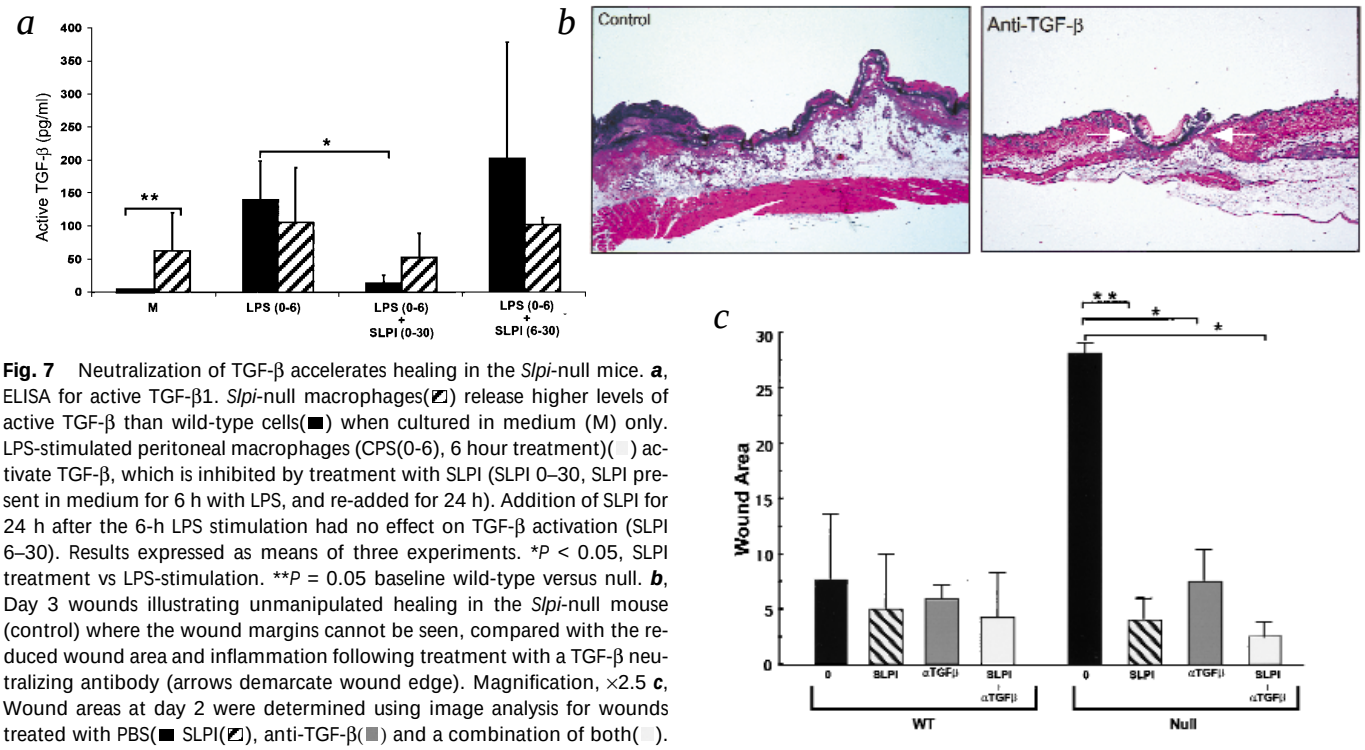


Fig. 7 Neutralization of TGF-β accelerates healing in the *Slpi*-null mice. ***a***, ELISA for active TGF-β1. *Slpi*-null macrophages(◩) release higher levels of active TGF-β than wild-type cells(■) when cultured in medium (M) only. LPS-stimulated peritoneal macrophages (CPS(0-6), 6 hour treatment)(□) activate TGF-β, which is inhibited by treatment with SLPI (SLPI 0–30, SLPI present in medium for 6 h with LPS, and re-added for 24 h). Addition of SLPI for 24 h after the 6-h LPS stimulation had no effect on TGF-β activation (SLPI 6–30). Results expressed as means of three experiments. *P < 0.05, SLPI treatment vs LPS-stimulation. **P = 0.05 baseline wild-type versus null. ***b***, Day 3 wounds illustrating unmanipulated healing in the *Slpi*-null mouse (control) where the wound margins cannot be seen, compared with the reduced wound area and inflammation following treatment with a TGF-β neutralizing antibody (arrows demarcate wound edge). Magnification, ×2.5 ***c***, Wound areas at day 2 were determined using image analysis for wounds treated with PBS(■ SLPI(◩), anti-TGF-β(▩) and a combination of both(□). Results are expressed as means ± s.d. *P < 0.05 compared with control wounds. **P = 0.007, *P = 0.03.

concomitantly with, LPS, no inhibition occurred if SLPI was added six hours after the cells were stimulated (Fig. 7*a*). These observations favor the idea that SLPI is acting at the level of macrophage activation and is not acting directly on cell-independent proteolysis. *Slpi*-null cells, even without LPS stimulation, constitutively activate greater quantities of TGF-β than their wild-type counterparts (Fig. 7*a*).

Neutralization of TGF-β *in vivo* reverses impaired healing

Because an absence of SLPI resulted in increased local levels of active TGF-β1, we reasoned that this potent chemoattractant was involved in the excessive inflammation and impaired healing phenotype observed. A reversal of the aberrant healing in null mice, including a decrease in the inflammatory response, could be achieved with neutralizing antibodies to TGF-β applied at the onset of healing (Fig. 7*b*,*c*; P = 0.03). Anti-TGF-β monotherapy was less effective than SLPI alone or together with anti-TGF-β (wound areas, 7.5 ± 2.8 for anti-TGF-β versus 2.5 ± 1.3 for SLPI plus anti-TGF-β; P = 0.03), suggesting that SLPI was the dominant mediator of enhanced healing, with additional effects (such as anti-inflammatory and anti-proteolytic actions) beyond its anti-TGF-β activity. By comparison, application of neutralizing antibodies to TGF-β to the rapidly healing wounds of wild-type mice did not accelerate the healing response (Fig. 7*c*). These data suggest that enhanced activation of latent TGF-β at the wound site in *Slpi*-null mice is a key mechanism responsible for the early influx of inflammatory cells, the latter acting not only as a source of proteases, but also of additional chemoattractants and inflammatory mediators.

Discussion

We have demonstrated using gene targeting that SLPI has an important role in cutaneous wound healing. Absence of SLPI leads to delayed wound healing, an increased and prolonged inflammatory response, enhanced elastase activity and delayed matrix accumulation. Thus, SLPI may have a number of diverse roles in the wound-healing cascade. Three major functions of SLPI can be delineated: inhibition of local elastase; controlling of leukocyte activation; and reduction of TGF-β activation, leading to a reduced inflammatory response. Previous studies have implicated the importance of raised elastase levels and prolonged inflammation in delayed wound-healing states, specifically chronic human wounds and acute wounds in the elderly[18,20,29,38]. Elastase is capable of degrading a wide variety of structural and functional proteins (such as proteoglycans, collagen and fibronectin) deposited in wounds[21,23]. This leads to reduced matrix deposition and impaired wound healing, generating a cascading effect, as elastase activates matrix metalloproteinases locally leading to amplified matrix cleavage[11]. Moreover, evidence suggests that SLPI inhibits synthesis of macrophage collagenase, likely through inhibition of NF-κB (refs. 10,11), implicating SLPI as a crucial molecule in the control of excessive tissue destruction. In agreement with this proposed role of SLPI in wound healing, the application of exogenous SLPI to the wounds reduced the local inflammatory response and enhanced the rate of healing.

A previously unrecognized link between SLPI and TGF-β has emerged from our studies. Although a variety of proteases may contribute to activation of TGF-β, the most potent chemoattractant for inflammatory cells *in vitro* and *in vivo*[25], serine proteases inhibitable by SLPI were not considered pivotal. Aside from suppressing elastolytic release of TGF-β complexed with elastin[35], however, SLPI appears to control TGF-β activity by regulating cellular activation. Macrophage cleavage of latent to active TGF-β is an important regulatory point in the bioactivity of TGF-β, requiring prior cellular activation[32]. We suggest that the increased activation of TGF-β locally, in the absence of SLPI, is integral to the early and sustained influx of inflammatory cells. The pool of available latent TGF-β, rapidly released at the site of injury by degranulating platelets and inflammatory cells, may also be increased secondary to the elastase-mediated release of LTBP from

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elastin fibers[34,39]. The spatial and temporal localization of SLPI supports its role as an anti-proteolytic mediator, as SLPI has been demonstrated in association with human dermal elastin fibers adjacent to proliferating keratinocytes[14]. Moreover, the elastin-derived peptides released secondary to elastase activity have chemoattractant properties for inflammatory cells, thus perpetuating the proteolytic milieu[40]. In the absence of SLPI, this balance between proteolytic and anti-proteolytic forces is disrupted, barring repair. The release of active TGF-β in this milieu appears to be a major contributing factor to aberrant wound repair. Notably, the neutralization of TGF-β in the early phases of wound healing reduces inflammation in the wounds of *Slpi*-null mice and reverses the abnormal wound-healing phenotype, suggesting that enhanced TGF-β activity in the absence of SLPI is crucial to the differences in healing observed. These data are counterintuitive to the dogma that increased TGF-β levels promote healing, but consistent with recent studies demonstrating that impaired TGF-β signaling enhances wound healing[38], and suggest that TGF-β antagonists may provide a new approach to certain types of pathologic tissue repair. These data re-affirm the concept that in the absence of infection, the inflammatory response is excessive and potentially detrimental to wound repair[38].

The functions of SLPI deduced from our studies may have potential therapeutic value in impaired wound healing states characterized by excessive elastase activity and matrix degradation[20]. For example, treatment of chronic wounds and delayed acute wounds in the elderly with SLPI may be beneficial in accelerating the wound-healing process. Moreover, our model of early uncontrolled inflammatory cell influx coupled with proteolysis is the first, to our knowledge, to mimic the major components of age-related wound healing. It is possible that these findings may be extended to other pathophysiological states involving excessive elastase activity, and also to those in which TGF-β initiates excessive inflammation and fibrosis[41,42].

The generation of this animal model, reflecting aspects of age-related acute and chronic wound healing in humans, makes it possible to experimentally elucidate critical aspects of these processes. Moreover, assessment of interventional strategies to inhibit the development of chronicity and to facilitate healing are key to alleviating the morbidity associated with non-healing wounds. The multiple functions of SLPI in innate host defense, including its anti-proteolytic, anti-microbial and anti-inflammatory profiles, suggest not only an important endogenous role, but also a therapeutic role in aberrant inflammation and infection. In this regard, the presence of SLPI in saliva and the perceived efficacy of licking of skin wounds may be a consequence of localized delivery of molecules such as SLPI, and may reflect an important evolutionary advantage with respect to the healing of wounds in nature.

Methods

Isolation of *Slpi* from 129/SvJ mouse. Total RNAs were isolated from C3H/HeJ mouse salivary gland and reverse transcribed with primers mP1 (5′–CACCATGAAGTCCTGCGG–3′; sense) and mP2 (5′–CCGAGCACGAGTCCAGAGCA–3′; antisense). The PCR product was further amplified with primers mP3 (5′–GGAGGCAAAAATGATGCTATCA–3′; sense) and mP4 (5′–GGCGCCAATGTCAGGGATCAG–3′; antisense) to generate a 347-bp *Slpi* cDNA fragment and confirmed by sequence analysis. The cDNA fragment was used as probe to screen a 129/SvJ genomic library (Stratagene, La Jolla, California). The mouse gene encoding SLPI was found to span 2.6 kb and is comprised of 4 exons and 3 introns.

Generation of *Slpi*-mutant mice. *Slpi*-null mice were generated by targeted disruption of *Slpi* by homologous recombination. A pPNT targeting vector was constructed in which the 849-bp promoter region and 46-bp exon I of mouse *Slpi* were replaced with a PGK-neomycin (neo) cassette containing a diagnostic *Sfu*1 site. The 16 residue truncated protein is inactive due to the absence of promoter and reading frame changes. A PGK-herpes simplex virus thymidine kinase (HAS-tk) cassette was included for negative selection. The construct was introduced into R1 ES cells and subjected to selection with G418 and ganciclovir. Southern blots of *Xba*I- and *Sfu*I-digested genomic DNA were hybridized with a 3′ flanking 313-bp probe to identify 3 targeted clones from 206 double-resistant clones. The primers 5′–GAATTCCAGGCTTGGGCATGCT–3′ (sense) and 5′–GGTAATAACCAGAAAGTGACAGAG–3′ (antisense) were used to clone the 313-bp fragment by PCR. The endogenous wild-type locus yielded the expected 12-kb *Xba*I band and the targeted ES locus yielded the expected 4.6-kb *Xba*I/*Sfu*I band with probe. Cells from targeted ES-cell clones were injected into blastocysts from C57BL/6 mice and transferred into B6D2F1 foster mothers. Two independent clones (E139 and E154) were used for germline transmission. Male chimeras were bred with C57BL/6 females, and the germline mice carrying the targeted allele and their F2 offspring were identified by either Southern hybridization using the 313-bp probe described above or PCR analysis of tail DNA. On Southern-blot analysis, wild-type allele yielded a 12-kb band and targeted allele yielded a 4.6-kb band. A specific primer pair (P1/P2) from the deleted promoter region was used to amplify a 490 bp fragment of wild-type allele. A neo-specific primer pair, neo1/neo2, was used to amplify a 344-bp fragment of the targeted allele. The primer sequences were as follows: N1, 5′–CGCTTCCTCGTGCTTTACGGTATC–3′; N2, 5′–GATGCCAGGCATTTGCACTGCCG–3′; P1, 5′–CAAGGCCTTCTGTGTGTAACTTTC–3′; P2 5′–CTGCTACAGAAGTAGGTGGCAGAC–3′.

Northern-blot hybridization. Total RNA was extracted from multiple organs including normal skin (day 0) and day 1, 2, 3, 5, 7 and 14 wounds. Total RNA (10 μg) was electrophoresed in a 1% agarose/formaldehyde gel, transferred to a nylon membrane (Turboblotter system) and probed with radiolabeled 340-bp mouse *Slpi* cDNA.

Wound-healing experiments. Eight-week male and female mice (SLPI wild-type and null) were anesthetized with inhaled methoxyfluorane, the dorsum shaved and cleaned with alcohol. Four equidistant 1cm full-thickness incisional wounds were made through the skin and panniculus carnosus muscle, and left to heal by secondary intention[38]. Wounds harvested were bisected for histology and immunostaining, or snap-frozen in liquid nitrogen. For a subset of animals, immediately before wounding (day 0), the area to be incised was injected intra-dermally with 50 μls of rat SLPI peptide[11] (1 μg/wound), PBS vehicle or left unmanipulated, and wounds were injected again at day 1. For another subset, the site to be incised was injected intra-dermally with PBS (50 μl), anti-TGF-β1/2, control antibody (1 μg/wound), combined SLPI (1 μg) and anti-TGF-β (1 μg) with one subsequent injection of SLPI (1 μg) at day 1 post-wounding, or left unmanipulated. Treatments were rotated to ensure no site bias. Heparinized blood was extracted by cardiac puncture for differential leukocyte count analysis.

Histology, immunocytochemistry and image analysis. Paraffin sections were stained with hematoxylin and eosin (H&E), Giemsa, Picrosirius Red (where staining intensity correlates to tissue hydroxyproline levels[43]), or immunohistochemistry with antibodies specific for rat SLPI (5 μg/ml; ref. 11), Mac2 (1.4 μg/ml) or collagen I (1:20, DAKO, Carpinteria, California). Primary antibody was detected using a FITC-labelled secondary (1:40 dilution for 20 min) or using the peroxidase kit as indicated (Vector, Burlingame, California). Immunostaining for active and latent TGF-β was performed using an antibody specific to the active form[31,44] (LC 1-30), and an antibody which recognizes total TGF-β raised to the C terminus of TGF-β (rabbit IgG, TGF-β1(V): sc-146; Santa Cruz Biotechnologies, Santa Cruz, California; ref. 26).

RNase protection and elastase assays. RNase Protection assays (RPAs) were performed using multiprobe templates (Pharmingen, San Diego, California) on total RNA (3 μg) from wild-type and null wounds, and developed using phosphorimaging. Elastase activity was determined as described[18,20].

Monocyte functional studies. Bone marrow was collected from mouse fe-

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murs and tibias, and mononuclear cells were isolated using a two-component step gradient[38] (Cardinal Associates, Santa Fe, California). For phagocytosis and superoxide radical production, FACS analysis was performed using freshly extracted bone marrow neutrophils and monocytes (Fc Oxyburst assay reagents, Molecular Probes, Eugene, Oregon). Cells were suspended in 4 °C glucose-containing, calcium and magnesium-free PBS and pre-warmed to 37 °C. Baseline readings of fluorescence intensity were taken and fluorescein-labeled immune complexes (20 μl; IC) added to 2×10^6 cells/ml to give a final concentration of 120 μg IC/ml. Readings were taken every 5 s for 2 min following addition of the ICs, which fluoresce when internalized and oxidized.

TGF-β activation by macrophages. Thioglycollate-elicited peritoneal macrophages were obtained from mice injected intraperitoneal (i.p.) with 4% thioglycollate (2 ml) 4 d before collection[32]. Macrophages were collected by peritoneal lavage with 2×5 ml of ice-cold DMEM. Cells were pelleted at 100*g* for 10 min, resuspended in complete medium, plated (2×10^5 cells) in 24-well plates and non-adherent cells removed after 3 h. Cells were pre-treated with media or rat SLPI (10 μg/ml) for 1 h before and during LPS stimulation (100 ng/ml) for 6 h. After 6 h, cells were washed and incubated in serum-free medium for an additional 24 h with or without SLPI (10 μg/ml). The concentration of active TGF-β in the medium was determined using an ELISA kit[45] (Promega, Madison, Wisconsin).

Statistical analysis. Statistical differences were determined using *t*-test or ANOVA. A P value of < 0.05 was considered significant.

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